Computational Exploration of Bond Order Calculations through Density Matrices

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Abstract

Bond order is a fundamental concept in chemistry that quantifies electron sharing between atoms, providing essential insights into molecular stability, reactivity, and electronic structure. While classical Lewis structures offer a simplified view of bonding, advanced computational methods are required to accurately describe complex, delocalized, or conjugated systems. This work presents a computational exploration of bond order formulations through a custom C++ implementation of the Self-Consistent Field (SCF) method. By iteratively solving for molecular electronic structures, the program generates the critical density and overlap matrices required for various bond order definitions.

This study implements and compares three primary formulations: the Wiberg, Mayer, and Mulliken methods. The Wiberg index utilizes squared density matrix elements in an orthogonalized basis to align with classical interpretations, while the Mayer method extends this approach to nonorthogonal basis sets by explicitly incorporating both density and overlap matrices. Additionally, the Mulliken method is evaluated, which partitions electron density through a symmetric split of the products of density and overlap matrix elements. These methodologies are applied across a diverse set of molecular systems, including simple diatomics, polar polyatomics, and triple-bond systems. The primary objective is to evaluate how differing mathematical treatments of orbital overlap and electron density influence the representation of chemical bonding and to identify the most appropriate formulation for specific computational and chemical scenarios.

The formulas we implemented were evaluated across a diverse set of molecular systems, including simple diatomics, polar polyatomics, and triple-bond systems. Results indicated that while all three methods show high agreement in single-bonded systems like H_2, HF, and H_2O, they diverge significantly in multiple-bond and conjugated environments. Specifically, the Wiberg method maintained stability in high-overlap systems such as HCN, whereas Mayer and Mulliken formulations exhibited substantial value inflation due to their sensitivity to non-orthogonal overlap. The study concludes that no single bond order definition is universally superior; the appropriate formulation must be selected based on specific electronic characteristics, such as polarity or degree of orbital overlap, to accurately represent chemical interactions within a computational framework.

Background

Bond order quantifies the number of electrons that are shared between two atoms. Knowing bond order allows scientists to understand the chemical structure of a molecule. While simple molecules can often be interpreted through classical Lewis structures, more advanced methods are required to describe bonding in delocalized, conjugated, or other complex systems. A quantum molecular orbital approach allows bond order to be derived directly from overlap matrices and electron density distributions, providing a more detailed representation of chemical interactions. One practical application of this approach is the ability to assess molecular stability, strength, and length.

Self-consistent field (SCF) methods provide an effective framework for studying molecular electronic structure by iteratively solving for the electron distribution within a molecule. Through repeated construction and diagonalization of the Fock matrix, SCF procedures generate molecular orbitals, density matrices, and overlap matrices that describe the electronic behavior of molecular systems. These matrices form the foundation for several bond order formulations designed to quantify electron sharing between atoms.

The most widely used bond order definitions are the Mulliken, Wiberg, and Mayer methods. The Wiberg bond simply squares the density matrix contributions in an orthogonalized basis, resulting in bond orders that correlate more closely with classical Lewis structure interpretations. Mayer bond orders extend Wiberg's theory further by explicitly incorporating both density and overlap matrices, extending the calculation to nonorthogonal basis sets. The Mulliken population analysis method estimates bond orders using the products of density and overlap matrix elements. The final result is multiplied by two to account for both spins of the electron.

This work presents a computational exploration of bond order calculations through density matrices using a custom C++ implementation of an SCF-based quantum chemistry method. The program was developed to generate overlap and density matrices and to evaluate chemical bonding using Mulliken, Wiberg, and Mayer bond order formulations. By comparing these methods across molecular systems, the study investigates how differing mathematical treatments of electron density and orbital overlap influence calculated bond order values. The project also demonstrates how quantum chemistry methods can be implemented computationally to provide insight into molecular structures and chemical bonding behavior.

Objectives

Using the density and overlap matrices generated through the SCF method, this work computes bond orders for a variety of molecular systems using C++. We compare three common bond order formulations: Wiberg, Mayer, and Mulliken bond order methods. The primary objective is to investigate how these different formulations describe electron sharing and chemical bonding in molecular systems. In particular, this research examines the strengths, limitations, and practical applications of each method to determine which bond order approach is most appropriate for specific chemical and computational scenarios.

More specific objectives of this project include:

1. Developing a C++ program capable of performing SCF calculations and generating molecular overlap and density matrices.
2. Implementing Mulliken, Wiberg, and Mayer bond order calculation methods within the program's framework.
3. Comparing the mathematical formulations and computational behavior of the Mulliken, Wiberg, and Mayer bond order approaches.
4. Evaluating how each bond order method represents bonding interactions across different molecular systems and determining how closely the calculated values align with expected chemical behavior.

Methodology

This project utilized a custom C++ implementation of the SCF method to investigate bond order calculations through density and overlap matrices. Molecular geometries for several molecular systems were used as input for the program. Using SCF overlap matrices, Hamiltonian matrices, and molecular orbital coefficients were generated to describe the electronic structure of each molecule.

Molecular orbital coefficients obtained from diagonalization of the Fock matrix were used to construct the density matrix according to Equation 1.

$$1)\quad P_{\mu\nu} = 2\sum_{i}^{occ} c_{\mu i} c_{\nu i}$$

The overlap matrix, which describes the overlap between basis functions, was computed using Equation 2.

$$2)\quad S_{\mu\nu} = \int \phi_{\mu}(r)\, \phi_{v}(r)\, dr$$

After convergence of the SCF procedure, bond orders were calculated using Mulliken, Wiberg, and Mayer equations. The Wiberg equation emphasizes electron sharing through squared density matrix contributions and often produces values that correlate closely with classical Lewis bonding descriptions. The Wiberg bond index was calculated using the density matrix in an orthogonalized basis according to Equation 3.

$$3)\quad B_{AB} = \sum_{\mu\in A}\sum_{\nu\in B} P_{\mu\nu}^{2}$$

The Mayer equation incorporates nonorthogonal overlap effects directly through the product of the density and overlap matrices, providing improved consistency across molecular systems. Mayer bond order was computed using both the density and overlap matrices as shown in Equation 4.

$$4)\quad B_{AB} = \sum_{\mu\in A}\sum_{\nu\in B} PS_{\mu\nu}\, PS_{\nu\mu}$$

Mulliken bond order was calculated using products of density and overlap matrix elements as shown in Equation 5. In this formulation, the summation is performed over atomic orbitals associated with atoms A and B. The factor of 2 accounts for electron spin pairing.

$$5)\quad B_{AB} = 2\sum_{\mu\in A}\sum_{\nu\in B} P_{\mu\nu} S_{\mu\nu}$$

Bond orders were compared by running a script that calculates the difference between expected standard Lewis chemistry bond orders (0, 1, 2, 3) and the calculated bond order. These differences for each value in the bond order matrix were then averaged and presented as the Mean Absolute Error. Bond orders were also compared between each method, averaging the value for each bonded pair and presented with a standard deviation range.

Results and Discussion

The performance of the implemented bond order formulations was initially evaluated by calculating the Mean Absolute Error (MAE) across the test set, using standard valence bond integers as a baseline for comparison.

Table 1. Mean Absolute Error (MAE) calculated between standard Lewis bond order values and calculated bond order values.

Molecule	Wiberg	Mayer	Mulliken
H_2	4.44×10^{-16}	0.378	0.320
HF	0.050	0.148	0.009
HO	0.025	0.235	0.084
H_2O	0.013	0.195	0.069
HCN	0.055	1.11	0.787
C_2H_4	0.346	0.725	0.362

This summary table highlights the inherent sensitivity of each mathematical method to molecular complexity. A prominent result is the near-perfect accuracy of the Wiberg method for H_2, which yields a mean difference of 4.44×10^{-16}. This level of precision occurs because the Wiberg index operates within an orthogonal basis, perfectly matching the zero-differential overlap assumption of the CNDO/2 method we implemented in this project.

As molecular complexity increases, going from simple diatomics like H_2 to hybridized systems like ethene C_2H_4, the deviations generally rise across all formulations. Notably, the Mulliken method outperforms others in describing the polar H–F bond with a minimal error of 0.009, while the Mayer method consistently exhibits the highest deviations. These deviations are particularly noticeable in multiple-bond systems like HCN, where the Mayer error exceeds 1.0, demonstrating how specific mathematical treatments of the density and overlap matrices diverge in high-density electronic environments.

Polar Systems: H_2O and HF

In the water molecule, the formulations provide distinct representations of a polar polyatomic system. The Wiberg index yields a value of 0.981, aligning closely with the traditional single-bond representation. In contrast, Mayer reports a higher value of 1.26, demonstrating how the explicit inclusion of the Overlap Matrix (*S*) emphasizes the degree of electron sharing or covalency within the bond. Mulliken provides an intermediate value of 1.10, capturing shared density in a polar environment through its 50/50 partitioning approach. Interestingly, while Wiberg and Mulliken indicate nearly zero interaction for the non-bonded H_1–H_2 pair (10^{-5} and 0.002, respectively), Mayer detects a 0.057 interaction. This suggests that basis set overlap in a non-orthogonal framework can project small amounts of density between atoms that are not formally bonded.

The analysis of HF represents the impact of high electronegativity on these calculations. For this highly polar diatomic, Mulliken gives a value of 0.991, suggesting that its symmetric 50-50 partitioning can effectively model the distribution in simple polar bonds. Wiberg provides a value of 0.950, operating purely on the density matrix in orthogonal space. Mayer yields 1.15, reflecting the substantial orbital overlap between hydrogen and fluorine. This indicates that the *PS* product of the density and overlap matrices produces values that diverge from simple integer models when atoms are in close proximity, effectively capturing bond strength rather than just bond count.

Multiple-Bond and Complex Systems: HCN and C_2H_4

HCN serves as our most complex test case, since it is a linear triple-bond system. Wiberg provides a value of 3.04 for the C≡N interaction. By utilizing the density matrix squared (P^2) in an orthogonal space, it is able to focus on the shared electron pairs without being influenced by the tight spatial overlap. Mayer yields a significantly higher value of 5.01, demonstrating the sensitivity of the *PS* product to the high overlap found in triple bonds. This value is more representative of the total degree of shared density within the quantum framework than a literal count of chemical bonds. Mulliken yields 4.65; however, because it assumes a symmetric distribution of shared electrons, it does not fully account for how the electronegative nitrogen polarizes the electron cloud in HCN.

In ethene, the symmetry and hybridization of the planar double-bond system further differentiate the methods. The C=C interaction highlights the divergence seen in HCN, where Mayer (0.725 error) and Mulliken (0.362 error) show higher values than the integer baseline compared to Wiberg (0.346 error). Finally, the hydroxyl radical (HO) results demonstrate the formulations' behavior in open-shell systems. Wiberg remains consistent at 0.975 for the O-H bond, while Mayer's inclusion of overlap results in 1.23, capturing the unique electronic environment of the radical through increased sensitivity to the density distribution.

Table 2. Comprehensive bond order analysis

Molecule	Bond	Mean	StdDev	Wiberg	Mayer	Mulliken
H_2	H-H	1.23	0.166	1.00	1.38	1.32
HF	H-F	1.03	0.085	0.950	1.15	0.991
H_2O	O-H	1.12	0.116	0.981	1.26	1.10
HO	O-H (Radical)	1.10	0.106	0.975	1.23	1.08
HCN	C≡N	4.23	0.856	3.04	5.01	4.65
HCN	C-H	1.50	0.411	0.936	1.89	1.68
C_2H_4	C=C	4.19	0.676	3.29	4.91	4.38
C_2H_4	C-H	0.840	0.724	0.148	1.84	0.532

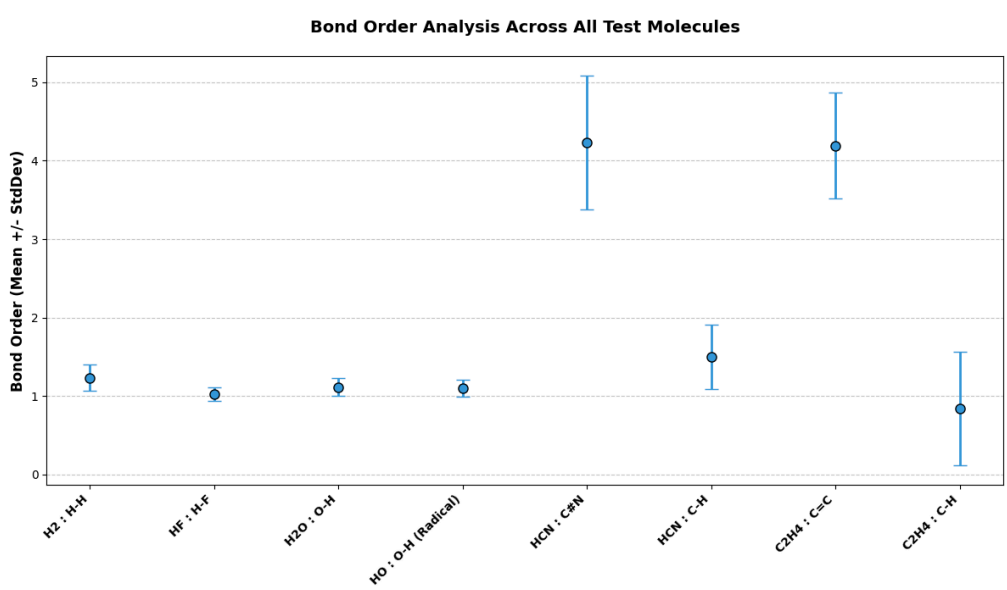

Figure 1. Comparative analysis of bond order formulations

Across the single-bonded molecules like HF, H_2O, and the HO radical, the three mathematical methods produce values that are very similar to one another. In these systems, the standard deviation stays between 0.085 and 0.116, showing that Wiberg, Mayer, and Mulliken all describe electron sharing in these environments in a very similar way. However, this overlap in results is not consistent in multiple-bond and conjugated systems like HCN and C_2H_4. In HCN, the standard deviation for the C≡N bond reaches 0.856, caused by a gap of nearly two units between the Wiberg (3.04) and Mayer (5.01) results. A similar split happens with the C-H bonds in ethene, which show a standard deviation of 0.724. These high variances represent a fundamental disagreement between the formulas; while orthogonal methods like Wiberg are not influenced by the physical overlap of atoms, non-orthogonal formulas like Mayer and Mulliken are highly sensitive to the dense electronic overlap found in π-systems, leading them to report a much higher level of shared density than the Wiberg method.

Conclusion

The results of this study indicate that no single formulation provides a universal bond order across all chemical scenarios, and we see that each method gives a unique perspective on the density and overlap matrices generated by the SCF procedure. The Wiberg method proved most reliable for systems where alignment with traditional methods and classical chemistry is required, though it is less descriptive in highly polar environments. The Mulliken method seems to be best suited for simple polar bonds where its symmetric partitioning effectively captures the distribution of shared electrons. Meanwhile, the Mayer method provides a deeper definition of bond strength and stability by incorporating non-orthogonal basis sets, making it more descriptive for complex systems despite its tendency toward higher numerical values in high-overlap environments. Ultimately, this work demonstrates that a computational approach to bond order requires selecting a mathematical formulation that aligns with the specific electronic characteristics of the target molecule. By implementing these three different methodologies into a custom C++ framework, we have highlighted how quantum chemistry can provide a detailed representation of chemical interactions.

Future Work

With more time, it would have been interesting to implement a 3-dimensional interactive visualization of the molecule's bond order, spatial representation, and electron density. We would implement this with VTK, a C++ 3-D rendering library. We would represent the molecule as a series of nodes, atoms, and edges, bonds. Each node's radius could be represented as the electron density size and the edge diameter could represent the bond order. This 3-D representation would provide a visual comparison of each bond order method.

It would also be interesting to further explore the application of the bond order. Since we read in a set of coordinates, it would be possible to move atoms away and towards each other. This is similar to potential energy testing for two molecules. A potential energy curve is created as seen in Figure 1. As the molecules get closer together, the potential energy is at the lowest to a point, and then the energy of the system rises again.

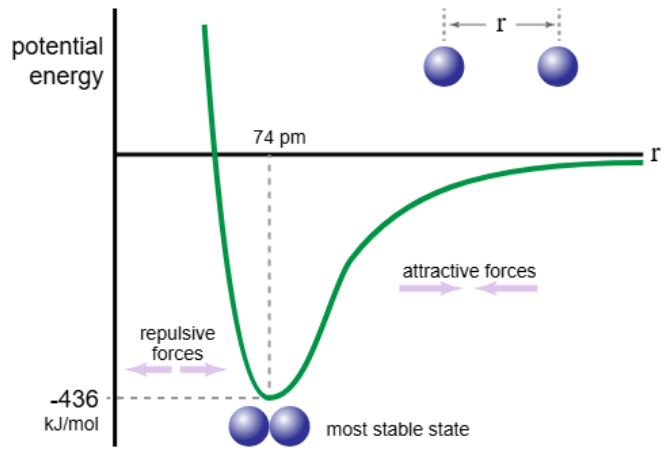

Figure 2. Potential energy curve. [University of Texas at Austin]

We hypothesize that we would see a similar pattern in the bond order, as it is representative of the shared electron density between two atoms. We would see a bond order at zero as the atoms are far apart, reach the true bond order as the optimal position is found, and approach zero again as bond stability is lost due to repulsive forces. Comparing each bond order formulation using this method could see how robust the or sensitive each approximation is to atom distances, adding another layer of testing to this research.

Team Contribution

Lily Hirano –

Created Github and base structure for overlap, density matrix calculations, and MO. Implemented environment, bond order calculations in C++, and tests in Python. Wrote background, objective, future work, and methodology. Cleaned up code and filled out ReadMe for submission..

Snigdha Chanduri –

Added files for and calculated values for HCN and C2H4. Wrote a script to obtain statistics for bond order calculations in C++. Created tables and graphs for bond order comparisons. Wrote abstract, results, and conclusion.

References

Wiberg

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Potential Energy Curve Diagram

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Supplementary

All code is available on GitHub at: https://github.com/lilyhirano/Molecule-Bond-Orders